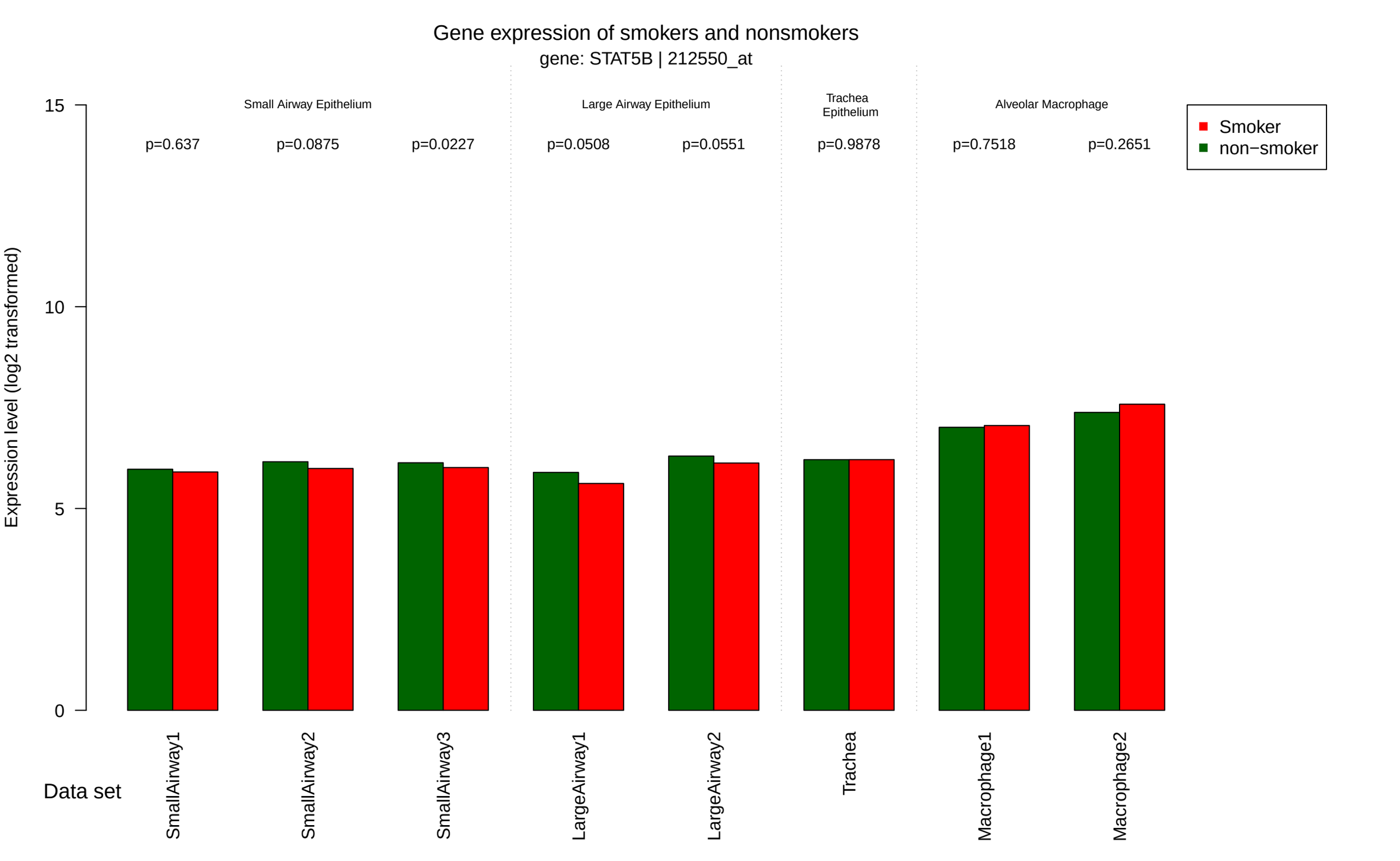
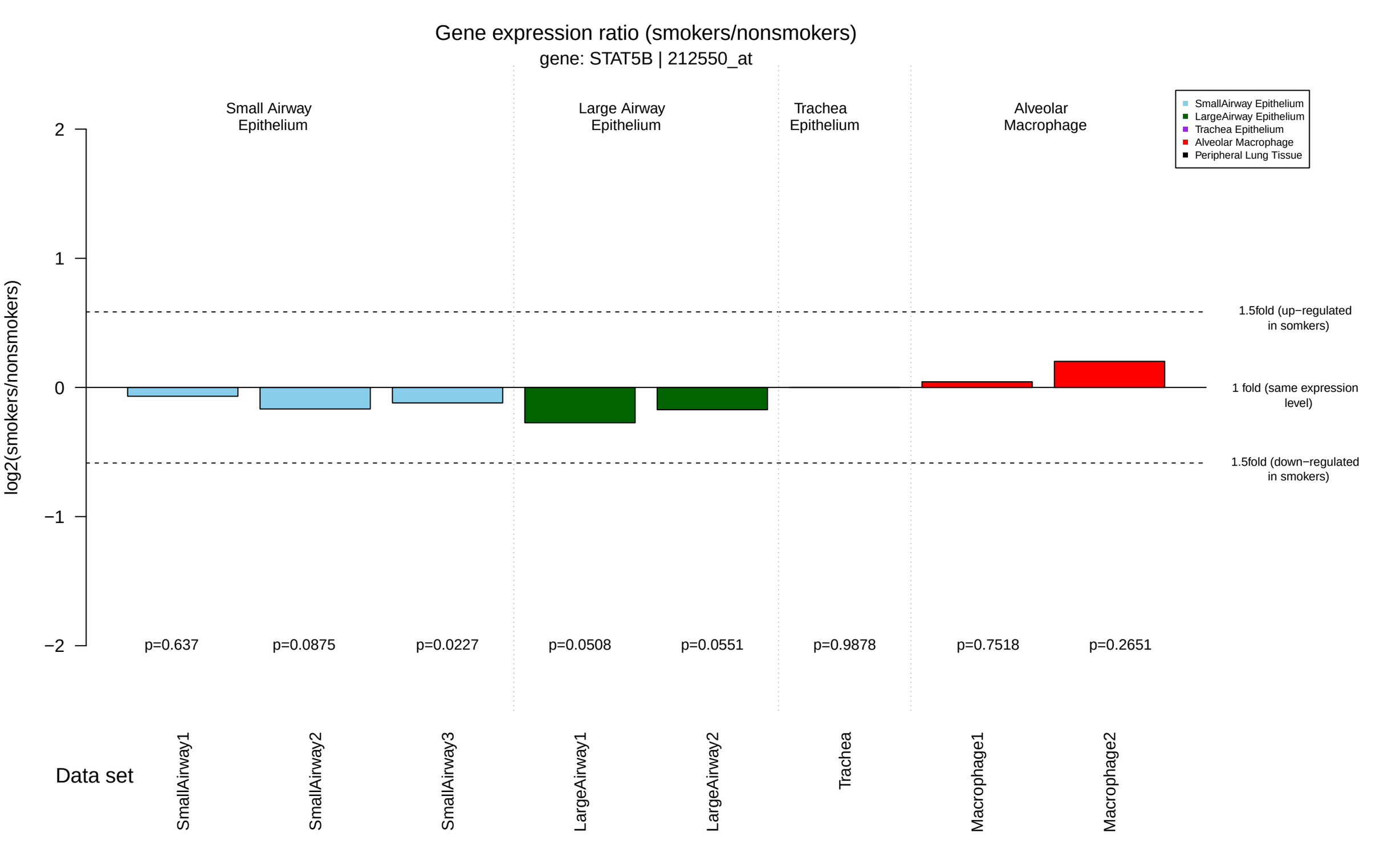
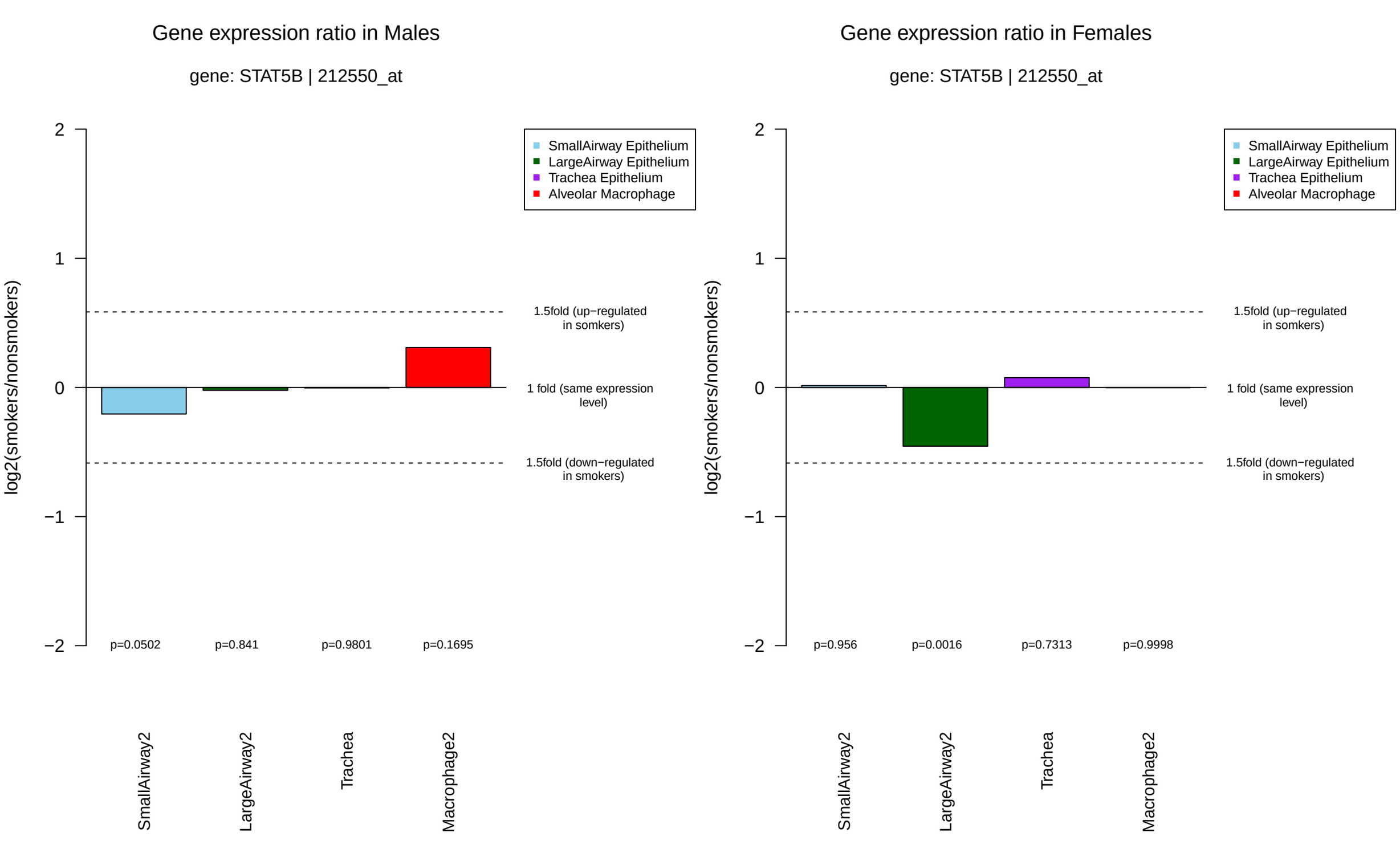
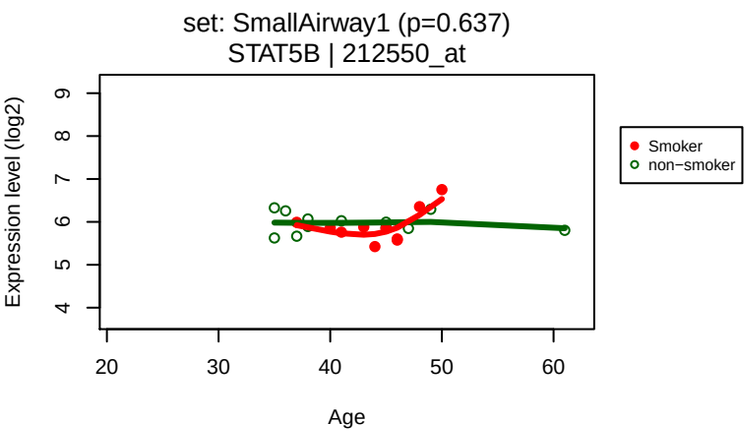
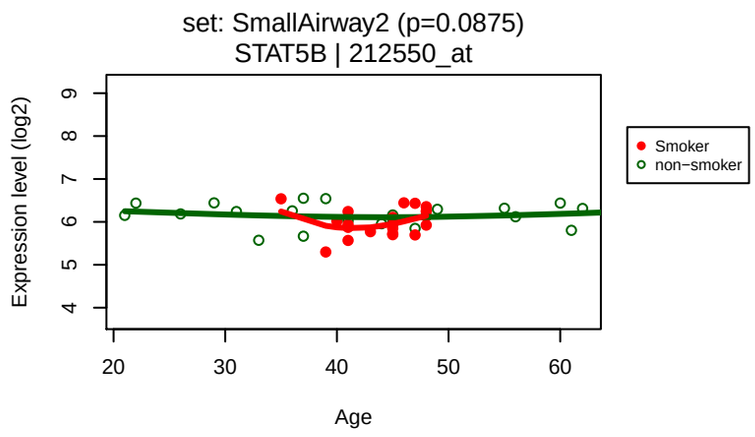
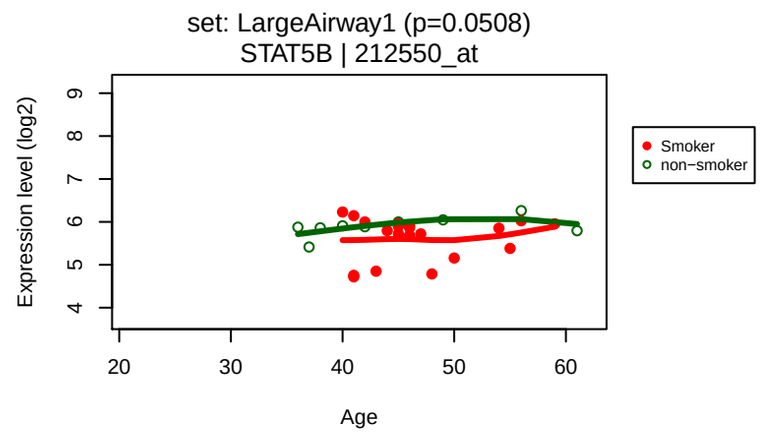

Gene expression of smokers and nonsmokers
gene: STAT5B | 212550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.637
p=0.0875
p=0.0227
p=0.0508
p=0.0551
p=0.9878
p=0.7518
p=0.2651
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

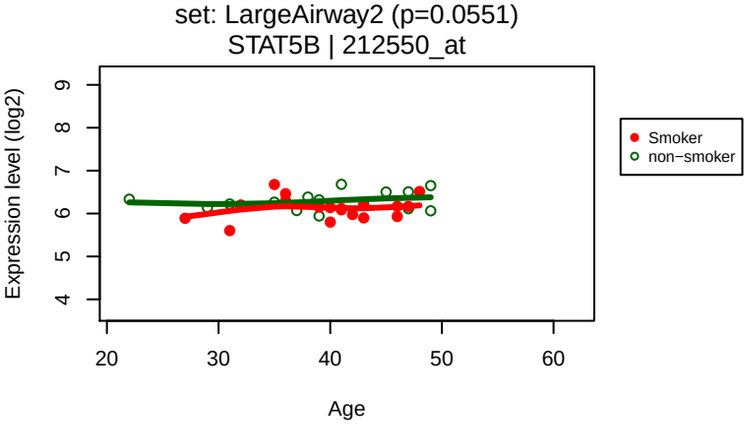

Gene expression ratio (smokers/nonsmokers)
gene: STAT5B | 212550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.637
p=0.0875
p=0.0227
p=0.0508
p=0.0551
p=0.9878
p=0.7518
p=0.2651
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

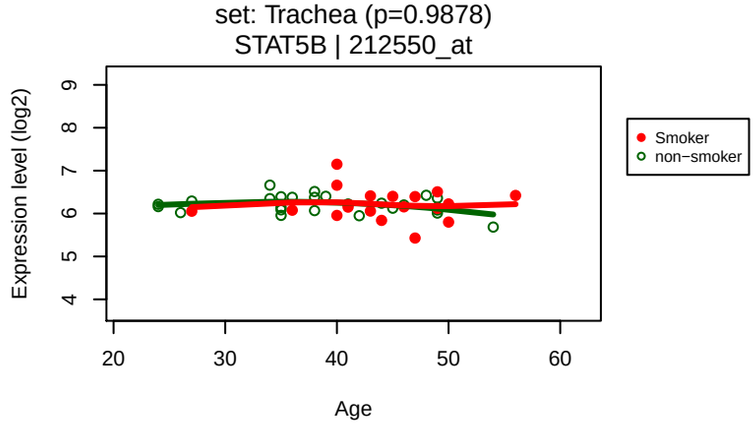

Gene expression ratio in Males
gene: STAT5B | 212550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0502
p=0.841
p=0.9801
p=0.1695
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT5B | 212550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.956
p=0.0016
p=0.7313
p=0.9998
SmallAirway2
LargeAirway2
Trachea
Macrophage2

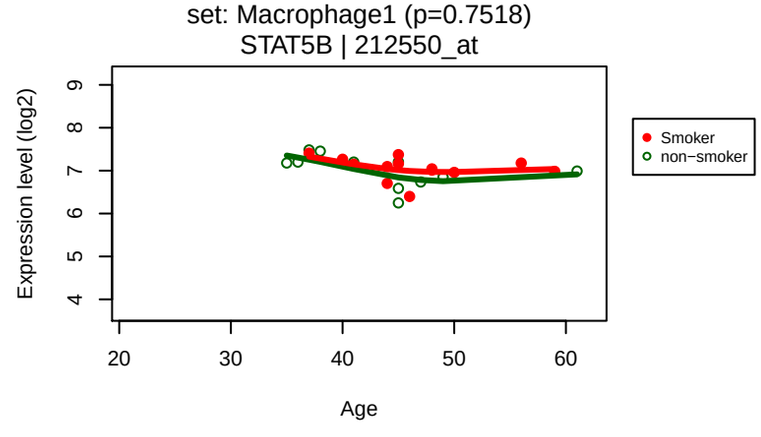

set: SmallAirway1 (p=0.637)
STAT5B | 212550_at
Expression level (log2)
Age
Smoker
non−smoker

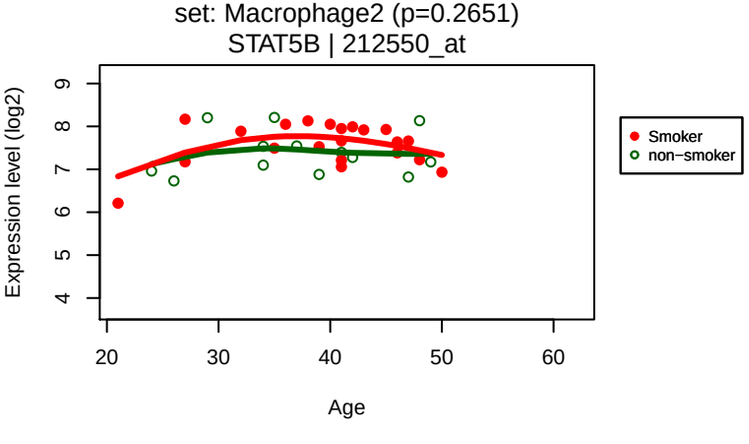

set: SmallAirway2 (p=0.0875)
STAT5B | 212550_at
Expression level (log2)
Age
Smoker
non−smoker

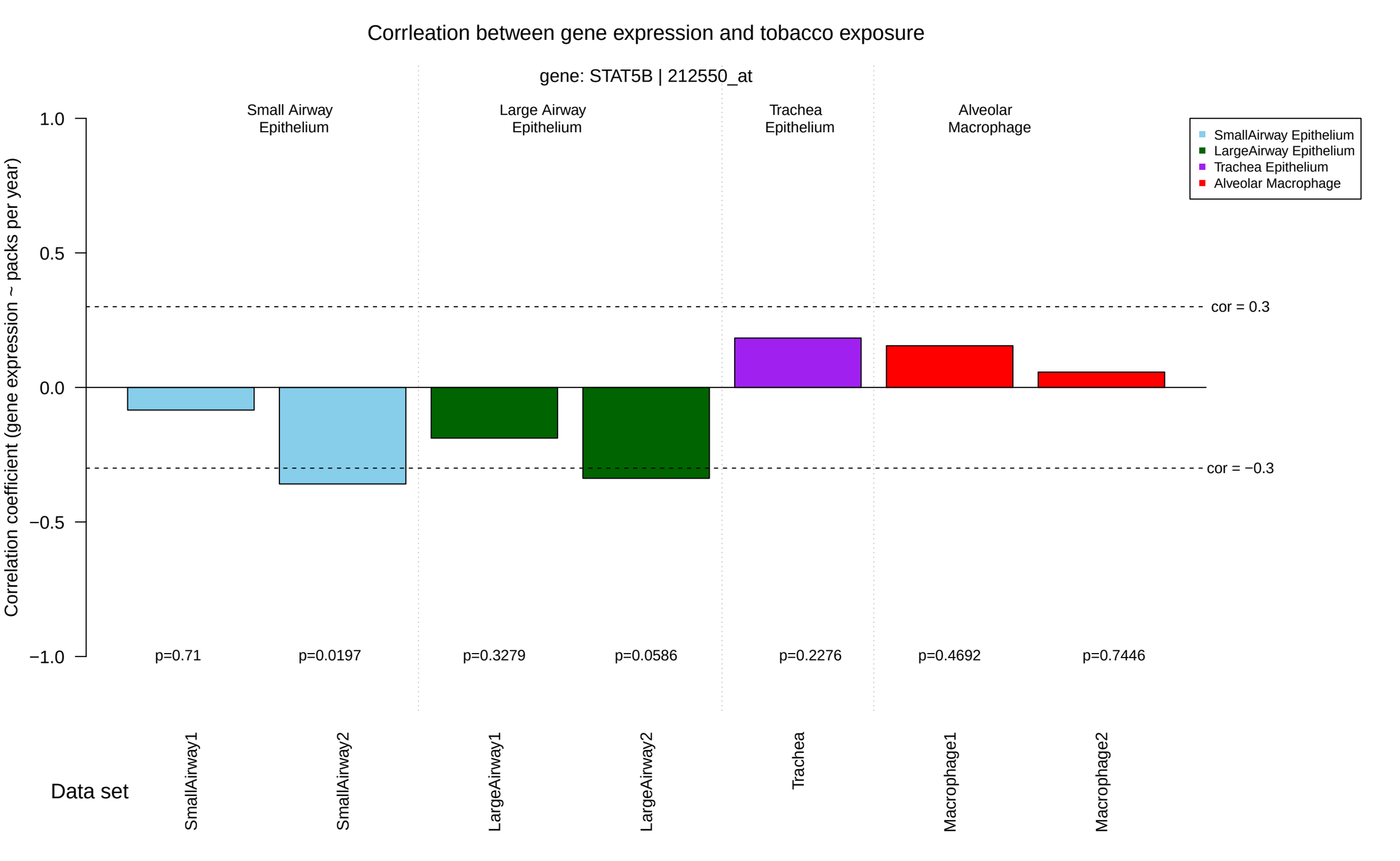

set: LargeAirway1 (p=0.0508)
STAT5B | 212550_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0551)
STAT5B | 212550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9878)
STAT5B | 212550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7518)
STAT5B | 212550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2651)
STAT5B | 212550_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT5B | 212550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.71
p=0.0197
p=0.3279
p=0.0586
p=0.2276
p=0.4692
p=0.7446
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2